Program: **Diploma in Microelectronics**	
Course Code: **4491**	Course Title**: Networks and Integrated Circuits**
Semester : **4**	Credits: **1**
Course Category: **Program Core**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **60**

Course Objectives:

- To develop the skill to diagnose and rectify the electric circuit networks
- To provide an insight on the fundamental concepts of operational amplifiers and their applications.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Engineering Mathematics principles and theorems	1002 2002	Mathematics I & I	1 & 2
Knowledge of current, voltage& magnetism	1003 2003	Applied Physics I & II	1 & 2
Representation of AC signals, voltage and power	2031	Fundamentals of electrical and electronic engineering	2
Design and working of Amplifiers and oscillators		Electronic Circuits	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Solve a given AC circuit to find various parameters using the concept of AC signals and the behavior of AC through various components.	13	Applying
CO2	Apply various network theorems for simplifying electric circuits and networks and illustrate the operations of transformers	15	Applying

CO3	Illustrate the internal structure, working and electrical parameters of operational amplifier.	15	Applying
CO4	Develop circuits using operational amplifier and timer IC	15	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	2						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Solve a given AC circuit to find various parameters using the concept of AC signals and the behavior of AC through various components.**		
M1.01	Illustrate the concept of AC signals	2	Understanding
M1.02	Explain the behavior of AC through R-L-C components	3	Understanding
M1.03	Solve the power in given AC circuits by keeping the concept of phasor diagram.	4	Applying
M1.04	Solve given series and parallel RLC circuits and find the various parameters	4	Applying
Contents: Concept of alternating voltage and current - Different waveforms (Sine, Square, Triangular and Sawtooth) - representation of alternating quantities - define the terms cycle, time period, frequency, amplitude, phase, maximum value, rms value, average value form factor AC through resistance, inductance, and capacitance (Solve simple problems) - Phasor diagrams- power factor definition - calculation of active, reactive and apparent power, Resonance and Q factor in an RLC circuit - series and parallel AC circuits (simple problems).			
CO2	**Apply various network theorems for simplifying electric circuits and networks and illustrate the operations of transformers.**		

<table>
<tr><td>M2.01</td><td>Illustrate various network theorems</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.02</td><td>Apply various network theorems for solving electrical and electronics circuits.</td><td>6</td><td>Applying</td></tr>
<tr><td>M2.03</td><td>Explain principle and operations of transformers</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.04</td><td>List the types and applications of Transformers.</td><td>3</td><td>Remembering</td></tr>
<tr><td></td><td>Series Test - I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents: Network Theorems and Transformers
Ohm’s law - Kirchhoff’s law – Mesh analysis – Node analysis- Superposition theorem - Thevenin's theorem - Maximum power transfer theorem (Solve simple problems)
Transformers: working principle of transformer - construction of transformer - elementary theory of an ideal transformer - voltage transformation ratio and rating of a transformer - emf equation derivation - losses in transformers -
types, applications of transformers</td></tr>
<tr><td>CO3</td><td colspan="3">Illustrate the internal structure, working and electrical parameters of operational amplifier.</td></tr>
<tr><td>M3.01</td><td>Illustrate the operation of differential amplifier</td><td>4</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Explain the block diagram of operational amplifier</td><td>2</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>Explain operational amplifier parameters</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>Illustrate the basic op-amp feedback circuits and develop various amplifier circuits.</td><td>6</td><td>Applying</td></tr>
<tr><td colspan="4">Contents:
Fundamentals of operational amplifier
Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics
Operational Amplifier: Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower - Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder – subtractor - Instrumentation amplifier.</td></tr>
<tr><td>CO4</td><td colspan="3">Develop circuits using operational amplifier and timer IC</td></tr>
<tr><td>M4.01</td><td>Develop wave-shaping circuits using op-amp</td><td>3</td><td>Applying</td></tr>
</table>

M4.02	Develop oscillator circuits using op-amp	3	Applying
M4.03	Illustrate comparator circuits using op-amp	3	Applying
M4.04	Develop active filters using op-amp	3	Applying
M4.05	Develop circuits using 555 timer IC	3	Applying
	Series Test – II	1	

Text/Reference

T/R	Book Title/Author
T1	B. L. Thereja and A. K. Thereja, "Textbook of Electrical Technology: Part 1 - Basic Electrical Engineering in S. I. Units", S. Chand Publication, 2012
T2	Roy D. C. and S. B. Jain, **Linear Integrated Circuits**, New Age International, 3/e, 2010
T3	Sergio Franco, **Design with operational amplifiers and analog integrated circuits**, 3rd Edition, Tata McGraw-Hill, 2008
R1	B. L. Thereja and A. K. Thereja, "Textbook of Electrical Technology: Part 2-AC and DC machines", S. Chand Publication, 2012
R2	Dr Ganesh Rao& R Sreenivasa, "Network Analysis: A simplified approach", Cengage
R3	Gayakwad R. A., **Op-Amps and Linear Integrated Circuits**, Prentice Hall, 4/e, 2010
R4	Botkar K. R., **Integrated Circuits**, 10/e, Khanna Publishers, 2010
R5	SalivahananS. ,V. S. K. Bhaaskaran, **Linear Integrated Circuits**, Tata McGraw Hill, 2008
R6	David A. Bell, **Operational Amplifiers & Linear ICs**, Oxford University Press, 2nd edition,2010

Online resources:

Sl.No.	Website Link
1	https://www.khanacadamy.org/electrical/

2	https://ocw.mit.edu/electrical
3	https://nptel.ac.in/courses/117107094/
4	https://nptel.ac.in/courses/117101106/
5	https://nptel.ac.in/courses/108/108/108108111/
6	https://inst.eecs.berkeley.edu/~ee140/fa19/
7	https://freevideolectures.com/course/2915/linear-integrated-circuits